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Research on Phenomenology of HNL and Possible Signals in Multi-Messenger Observation

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Recently, the IceCube detector has identified AGN NGC 1068 as a point source of high-energy neutrinos, while the γ ray observation exceeds the prediction of electromagnetic cascade models. This suggests some of the γ ray energy can be transported outside from being attenuated. In this paper we propose HNL, Heavy Neutral Leptons, as a way to explain this observation. We first review the origin of HNL theory, calculating the phenomenology behavior of certain processes related to $p-\gamma$ and $p-p$ neutrino production, then use it to try to explain the γ ray observation of AGN NGC 1068, and several other astronomical observations.

Keywords: HNL; multi-messenger; NGC 1068

1. Introduction

As advancements are made in astronomical observations, different types of new signals have became observable, along side with gravitational waves and wider range of electro-magnetic signals, neutrino signals are now observed by IceCube, which has recently spotted AGN NGC 1068 as a point source of neutrino flux around energy of 1 TeV,[1] and several other experiments, such as KM3NeT/ARCA, while still under construction, has observed a neutrino signal which possibly has energy as high as 220 PeV.[2]

Further more, neutrino has always been the strange one in Standard Model (SM), because of its tiny mass, thus weak coupling with the Higgs field, is considered some what unnatural. This leads neutrino and the origin of its small mass to be considered as a key to Beyond Standard Model Physics (BSM), such as Heavy Neutral Lepton (HNL) being candidate of dark mater. A lot of experiments on the earth is working on finding possible signals coming from BSM theories.[3] Although no definite signal has been observed, they do provide a solid constrain on the parameter space of possible BSM theories, for HNL with $3\,\mathrm{MeV} < m_N < 19.5\,\mathrm{MeV}$, the mixing angle with muon neutrino is constrained roughly as $|U_{\mu N}|^2 \lesssim 10^{-4}$.[4]

And, in the observation of NGC 1068 done by IceCube, when comparing with theoretical predictions[5] based on corona cascade models, exceeds the prediction of observed γ-ray flux around 10 GeV energy. This suggests that the hidden core of NGC 1068 has some process to transport more, electro-magnetic energy out. Axion is one option,[6] another option is HNL, which is discussed in this paper. Before NGC 1068, there are already attempts to combine multi-messenger observations in favor of constraining BSM theories.[7–9]

This paper is structured as follows. In Section , we outline the basic of HNL theory originating from neutrino mass, and calculate its phenomenology properties, while taking polarization into account. In Section , we investigate the multi-messenger observation results of NGC 1068, try to explain the exceeding γ-ray flux. Conclusion is in Section .

2. ISS Model

To explain the tiny mass of neutrino, a bunch of models known as Seesaw Mechanisms are proposed. For Type-I Seesaw model, we have

$$m_\nu = \frac{m_D^2}{M_R},\ m_N = M_R,\ |U|^2 = \left(\frac{m_D}{M_R}\right)^2 \tag{1}$$

Where m_D is the Dirac mass of neutrino, and M_R is the Majorana mass of the added right-handed neutrino field. For a sufficiently natural m_D and m_R, the mixing angle $|U|^2$ is always too small which makes observation impossible. How every by introducing another fermionic field, the mass matrix becomes

$$\left(\overline{\nu_L}\ \overline{\nu_R^c}\ \overline{S^c}\right) \begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & M \\ 0 & M & \mu \end{pmatrix} \begin{pmatrix} \nu_L^c \\ \nu_R \\ S \end{pmatrix} \tag{2}$$

The new parameter μ stands for potential lepton number violation, which should be small. The basic properties of neutrino and a pseudo-dirac fermion pair, which can be treated as a fermion because of the small μ, are

$$m_\nu = \mu\frac{m_D^2}{M^2},\ m_N = M,\ |U|^2 = \frac{m_\nu}{\mu} \tag{3}$$

3. HNL Production

Such HNL particles only interact through its mixed part of SM neutrino, which means those process which produces SM neutrino, if not prohibited by phase space, will also produce HNL at the same time.

3.1. *Pion decay*

Under the condition of AGN, $p-p$ and $p-\gamma$ processes are though to be the origin of high energy cosmic rays.[10] Those processes creates a large number of pions, and the decay of high-energy pions turns into high-energy cosmic rays, including neutrinos, γ-rays.[11]

After introducing HNL particle mixing with SM neutrino, charged pions can decay into HNL particles, intermediated by W bosons. By slightly modifying the

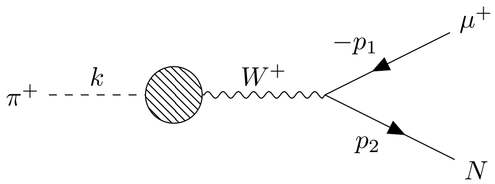

Fig. 1. Feynman diagram of π^+ decaying into μ^+ and massive HNL particle.

process of SM pion decay, we find that the decay width to HNL is

$$\frac{\Gamma_{\pi\to\mu N}}{\Gamma_{\pi\to\mu\nu}} = |U_\mu|^2 \times \rho_\pi(m_N) \tag{4}$$

Where

$$\rho_\pi(m_N) = \frac{(m_\pi^2(m_\mu^2+m_N^2) - (m_\mu^2-m_N^2)^2)\sqrt{\lambda(m_\pi^2,m_\mu^2,m_N^2)}}{m_\mu^2(m_\pi^2-m_\mu^2)^2} \tag{5}$$

$$\lambda(x,y,z) = x^2+y^2+z^2-2xy-2yz-2zx \tag{6}$$

It is useful to determine the helicity of the HNL particle, since it will play a role in the angular dependence when working on the decay of HNL. In the pion rest frame we have

$$\Gamma_{(s)} \sim (m_\mu^2+m_N^2)m_\pi^2 - (m_\mu^2-m_N^2)^2 - s(m_\mu^2-m_N^2)\sqrt{\lambda(m_\pi^2,m_\mu^2,m_N^2)} \tag{7}$$

Where s is the helicity of HNL, and $s=+1$ for right-handed polarized HNL particle. There are no differences in angular dependence for different helicity end states, since pion is unpolarized. After boosting to the lab frame, we must account for the polarization change on the HNL particles, thus for a pion spectrum $J(E_\pi)$, we get

$$Q_N^{(s)}(E_N) = \int_{\frac{E_N}{\lambda_{max}}}^{\infty} J(E_\pi)|U_\mu|^2\rho_\pi \times r_{(s)}(E_\pi,E_N) \times \frac{dE_\pi}{\lambda_{max}E_\pi} \tag{8}$$

Where

$$\lambda_{max} = \frac{E_{N,max}}{E_\pi} = \frac{m_\pi^2-m_\mu^2+m_N^2+\sqrt{\lambda(m_\pi^2,m_\mu^2,m_N^2)}}{2m_\pi^2} \tag{9}$$

$$\cos\theta_r = \frac{E_N(m_\pi^2-m_\mu^2+m_N^2)-2E_\pi m_N^2}{\sqrt{(E_N^2-m_N^2)\,\lambda(m_\pi^2,m_\mu^2,m_N^2)}} \tag{10}$$

$$r_{(s)}(E_\pi,E_N) = \frac{\Gamma_{(s)}\frac{1+\cos\theta_r}{2}+\Gamma_{(-s)}\frac{1-\cos\theta_r}{2}}{\Gamma_{(s)}+\Gamma_{(-s)}} \tag{11}$$

Here we assume $E_\pi \gg m_\pi$ to simplify the boost process.

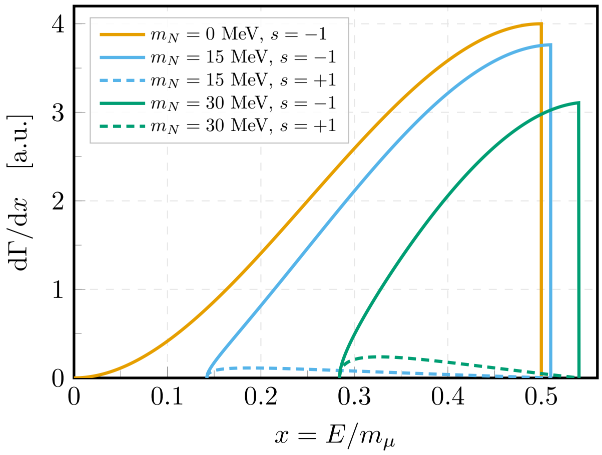

Fig. 2. Energy spectrum of HNL when muons decay into HNLs in muon rest frame. Normalized under $m_N = 0$, which basically resembles SM neutrino, with no right-handed product exists. When m_N increases, shape of the spectrum stays roughly the same, but the max energy increases while the total decay width decreases, and more right-handed HNLs, though still a small fraction, will be produced.

3.2. *Muon decay*

Muons will decay into HNL particles through a similar process as they decay into ν_μ. We find the decay width of $\mu^- \to Ne^-\bar{\nu}_e$ has the same form as the process of τ decay which has been studied,[12] since this process is symmetric between μ and τ, which is

$$\frac{\Gamma_{\mu\to N}}{\Gamma_{\mu\to\nu_\mu}} = |U_\mu|^2 \times \rho_\mu(m_N) \tag{12}$$

Where

$$\rho_\mu(m_N) = -r_N^8 + 8r_N^6 - 24r_N^4 \ln(r_N) - 8r_N^2 + 1, \quad r_N = m_N/m_\mu \tag{13}$$

And the spectrum of HNL particle under muon rest frame, separated by polarization, is

$$\frac{d\Gamma}{dE} = \frac{G_F^2}{24\pi^3}|U_\mu|^2 \times |\mathbf{p}| \cdot [3(m_\mu^2 + m_N^2)E - 4m_\mu E^2 - 2m_N^2 m_\mu - s \cdot |\mathbf{p}|(m_N^2 + 3m_\mu^2 - 4m_\mu E)] \tag{14}$$

Several examples are plotted in Fig. 2.

Though one can keep the muon polarization and calculate the angular dependence of how many different helicity HNL particles will be produced while the angle respect to muon polarization changes, we will simply ignore the muon polarization here as we assume that the magnetic environment of AGN will wipe the polarization out.

For a given muon spectrum, we have

$$Q_N^{(s)}(E_N) = \int_E^{+\infty} dE_\mu\, Q_\mu(E_\mu)|U_\mu|^2\rho_\mu \cdot \int_{x^-}^{x^+} \frac{dx}{E_\mu\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \cdot \left(\frac{1+\cos\theta_r}{2} g_N^{(s)}(x) + \frac{1-\cos\theta_r}{2} g_N^{(-s)}(x)\right) \tag{15}$$

Where

$$x^+ = \frac{m_N^2 + m_\mu^2}{2m_\mu^2},\ x^- = \frac{1}{2}\left(\frac{E_N}{E_\mu} + \frac{E_\mu}{E_N}\cdot\frac{m_N^2}{m_\mu^2}\right) \tag{16}$$

$$\cos\theta_0 = \frac{1}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}}\left(\frac{E_N}{E_\mu} - x\right) \tag{17}$$

$$\cos\theta_r = \frac{E_\mu}{\sqrt{E_N^2 - m_N^2}}\left(\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}} + x\cos\theta_0\right) \tag{18}$$

While $g_N^{(s)}(x)$ is normalized as

$$\int_{x^-}^{x^+} dx \sum_{s=\pm1} g_N^{(s)}(x) = 1 \tag{19}$$

4. HNL Decay

Straight-forwardly, HNL particles under ISS model will , and will only, decay into other leptons. Possible products are $\nu_\mu e^- e^+$ and $\nu_\mu \nu_l \bar{\nu}_l$. These processes are well studied,[12] we can recall the decay width for each process as

$$\Gamma_{N\to3\nu} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_\mu|^2 \tag{20}$$

$$\Gamma_{N\to\nu e\bar{e}} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_\mu|^2 \times C_1^e \tag{21}$$

Where

$$C_1^e = \frac{1}{4}\left(1 - 4\sin^2\theta_W + 8\sin^4\theta_W\right) \tag{22}$$

To simplify the problem we will ignore electron mass, and 3ν end state has taken all different flavor combinations into account.

Thus the branch ratio of HNL particle processing a visible decay is

$$BR_{N\to\nu e\bar{e}} = \frac{C_1^e}{1 + C_1^e} \approx 0.11 \tag{23}$$

The energy loss is pretty significant. Only one tenth of HNL decays will be observable. The angular and energy distributions of electrons/positrons are

$$
\begin{aligned}
g_e^{(1)}(x,\cos\theta) =& \frac{8}{C_1^e}\cdot x^2\cdot\{[(4-16s_W^2+64s_W^4)x-(1-4s_W^2+28s_W^4)]\cos\theta \\
&-(4-16s_W^2+64s_W^4)x+(3-12s_W^2+36s_W^4)\} \qquad (24)\\
g_e^{(2)}(x,\cos\theta) =& \frac{16}{C_1^e}\cdot x^2\cdot\{[(6-24s_W^2+32s_W^4)x-(3-12s_W^2+14s_W^4)]\cos\theta \\
&-(6-24s_W^2+32s_W^4)x+(3-12s_W^2+18s_W^4)\} \qquad (25)
\end{aligned}
$$

Where $x = E_e/m_N$, $s_W = \sin\theta_W$. θ is the angle between the out-going electron and initial HNL spin.

To get the total electron spectrum from a given HNL spectrum, we integral over the HNL spectrum and get

$$
\begin{aligned}
Q_e^{(Lab)}(E) = BR_{N\to\nu e\bar{e}}\cdot\int_E^{+\infty} dE_N\, Q_N^{(s)}(E_N) \\
\cdot\int_{\frac{E}{2E_N}}^{\frac{1}{2}}\frac{dx}{xE_N}\sum_{i=1,2} g_e^{(i)}(x, s(\frac{E}{xE_N}-1)) \qquad (26)
\end{aligned}
$$

Combining with 24 and 25, the remaining integral is

$$
\int_E^{+\infty}\frac{dE_N}{E_N}\,Q_N^{(s)}(E_N)[16y^3-27y^2+11-s\cdot(32y^3-63y^2+36y-5)] \qquad (27)
$$

Where $y = E/E_N$, and θ_W no-longer appears in this formula.

5. Multi-Messenger Observation of NGC 1068

6. General Appearance

Contributions to *International Journal of Modern Physics A* are to be in American English. Authors are encouraged to have their contribution checked for grammar. American spelling should be used. Abbreviations are allowed but should be spelled out in full when first used. Integers ten and below are to be spelled out. Italicize foreign language phrases (e.g. Latin, French). Upon acceptance, authors are required to submit their data source file including postscript files for figures.

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Major headings should be typeset in boldface with the first letter of important words capitalized.

8.1. *Subheadings*

Subheadings should be typeset in boldface italic and capitalize the first letter of the first word only. Section numbers to be in boldface roman.

8.1.1. *Subsubheadings*

Typeset subsubheadings in medium face italic and capitalize the first letter of the first word only. Section numbers to be in roman.

8.2. *Numbering and spacing*

Sections, subsections and subsubsections are numbered in Arabic. Use double spacing before all section headings, and single spacing after section headings. Flush left all paragraphs that follow after section headings.

8.3. *Lists of items*

Lists may be laid out with each item marked by a bullet:

- item one,
- item two.

Items may also be numbered in lowercase roman numerals:

(i) item one,
(ii) item two.
   (a) Lists within lists can be numbered with lowercase roman letters,
   (b) second item.

9. Equations

Displayed equations should be numbered consecutively in each section, with the number set flush right and enclosed in parentheses

$$\mu(n,t) = \frac{\sum_{i=1}^{\infty} 1(d_i < t, N(d_i) = n)}{\int_{\sigma=0}^{t} 1(N(\sigma) = n) d\sigma} \,. \tag{28}$$

Equations should be referred to in abbreviated form, e.g. “Eq. (28)” or “(28)”. In multiple-line equations, the number should be given on the last line.

For multiline equations, prefer using `align` or `align*` instead of `eqnarray`.

$$\begin{aligned} g(x) &= \sum_{k=1}^{N} y_k g(x_k) \\ &= \sum_{k=1}^{N} y_k \sum_{l=1}^{M} b_{kl} w_l \\ &= \sum_{k=1}^{N} \sum_{l=1}^{M} b_{kl} y_k w_l \end{aligned} \tag{29}$$

Displayed equations are to be centered on the page width. Standard English letters like x are to appear as x (italicized) in the text if they are used as mathematical symbols. Punctuation marks are used at the end of equations as if they appeared directly in the text.

10. Theorem Environments

Theorem 10.1. *Theorems, lemmas, propositions, corollaries are to be numbered consecutively in the paper or in each section. Use italic for the body and upper and lower case boldface, for the declaration.*

Remark 10.1. Remarks, examples, definitions are to be numbered consecutively in the paper or in each section. Use roman for the body and upper and lower case boldface, for the declaration.

Proof. The word 'Proof' should be typed in boldface. Proofs should end with a box. □

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Figures are to be embedded in the text nearest their first reference and sequentially numbered in Arabic numerals. The caption must be placed below the figure (see Fig. 3) and typeset in 8 pt roman with baselineskip of 10 pt. Use double spacing between a caption and the text that follows immediately.

Figures should be black and white (or half-tone) and of a good resolution and sharpness. Compressed formats, such as JPEG, should not use heavy compression, which introduces undesirable artifacts. Bitmap figures, such as photographs or half-tone images, should be at least 300 dpi (TIFF, BMP or JPEG). Vector (line-art) figures, such as charts and technical drawings, should be 600–1200 dpi (EPS, PS, PDF). The native formats of software packages will not be permitted.

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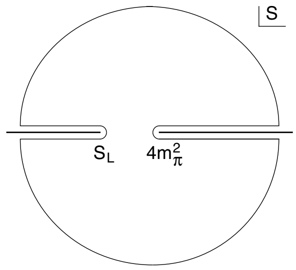

Fig. 3. A schematic illustration of dissociative recombination. The direct mechanism, $4m_\pi^2$ is initiated when the molecular ion S_L captures an electron with kinetic energy.

12. Tables

Tables should be inserted in the text as close to the point of reference as possible. Some space should be left above and below the table.

Tables should be numbered sequentially in the text in Arabic numerals. Captions are to be centralized above the tables (see Table 1). Typeset tables and captions in 8 pt roman with baselineskip of 10 pt.

Table 1. Comparison of acoustic for frequencies for piston-cylinder problem.

Piston mass	Analytical frequency (Rad/s)	TRIA6-S_1 model (Rad/s)	% Error
1.0	281.0	280.81	0.07
0.1	876.0	875.74	0.03
0.01	2441.0	2441.0	0.0
0.001	4130.0	4129.3	0.16

If tables need to extend over to a second page, the continuation of the table should be preceded by a caption, e.g. "*Table 2.* (*Continued*)".

13. Footnotes

Footnotes should be numbered sequentially in superscript lowercase roman letters.[a]

Acknowledgments

This section should come before the References. Dedications and funding information may also be included here.

[a]Footnotes should be typeset in 8 pt roman at the bottom of the page.

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Appendix A. Appendices

Appendices should be used only when absolutely necessary. They should come before the References. If there is more than one appendix, number them alphabetically. Number displayed equations occurring in the Appendix in this way, e.g. (A.1), (A.2), etc.

$$\begin{aligned} g_{\mu_1\mu_2} &= g_{axby} = -\epsilon_{abc}4\pi\,\frac{(x-y)^c}{|x-y|^3}\,, \\ h_{\mu_1\mu_2\mu_3} &= \epsilon^{\alpha_1\alpha_2\alpha_3} g_{\mu_1\alpha_1} g_{\mu_2\alpha_2} g_{\mu_3\alpha_3} \end{aligned} \tag{A.1}$$

with

$$\epsilon^{\alpha_1\alpha_2\alpha_3} = \epsilon^{b_1y_1b_2y_2cx} = \epsilon^{b_1b_2c}\delta(x-y_1)\delta(x-y_2)\,. \tag{A.2}$$

References

1. I. Collaboration*†, R. Abbasi, M. Ackermann, J. Adams, J. Aguilar, M. Ahlers, M. Ahrens, J. Alameddine, C. Alispach, A. Alves Jr *et al.*, *Science* **378**, 538 (2022).
2. *Nature* **638**, 376 (2025).
3. S. N. Gninenko, *Phys. Rev. Lett.* **103**, 241802 (Dec 2009), doi:10.1103/PhysRevLett.103.241802.
4. D. Bryman and R. Shrock, *Physical Review D* **100**, 073011 (2019).
5. K. Murase, S. S. Kimura and P. Meszaros, *Physical review letters* **125**, 011101 (2020).
6. B. P. Pant, *Physical Review D* **109**, 063002 (2024).
7. K. Hirata, T. Kajita, M. Koshiba, M. Nakahata, Y. Oyama, N. Sato, A. Suzuki, M. Takita, Y. Totsuka, T. Kifune, T. Suda, K. Takahashi, T. Tanimori, K. Miyano, M. Yamada, E. W. Beier, L. R. Feldscher, S. B. Kim, A. K. Mann, F. M. Newcomer, R. Van, W. Zhang and B. G. Cortez, *Phys. Rev. Lett.* **58**, 1490 (Apr 1987), doi:10.1103/PhysRevLett.58.1490.
8. T. Fischer, S. Chakraborty, M. Giannotti, A. Mirizzi, A. Payez and A. Ringwald, *Phys. Rev. D* **94**, 085012 (Oct 2016), doi:10.1103/PhysRevD.94.085012.
9. G. Magill, R. Plestid, M. Pospelov and Y.-D. Tsai, *Physical Review D* **98** (December 2018), doi:10.1103/physrevd.98.115015.
10. K. Murase and F. W. Stecker, High-energy neutrinos from active galactic nuclei, in *The Encyclopedia of Cosmology*, (WORLD SCIENTIFIC, October 2023), p. 483–540.
11. S. R. Kelner, F. A. Aharonian and V. V. Bugayov, *Physical Review D—Particles, Fields, Gravitation, and Cosmology* **74**, 034018 (2006).
12. K. Bondarenko, A. Boyarsky, D. Gorbunov and O. Ruchayskiy, *Journal of High Energy Physics* **2018** (November 2018), doi:10.1007/jhep11(2018)032.